Report — Assignment 7

Student: Hima Sai Chandana | Roll No: 2401MC37

1. Q1 — Kernel-Supported Threads (clone / join)

Design & Results

- clone(): Creates a thread that shares the parent's memory address space (pgdir) but gets its own kernel and user stacks. Execution starts at fcn(arg).
- join(): Waits for child threads to finish and cleans up their resources without freeing the shared memory.
- Result: Running 3 threads to increment a shared counter 100,000 times causes race conditions, leading to a final counter value less than 300,000.

2. Q2 — Counting Semaphores & Synchronization

Design & Results

- Kernel Semaphores: Implemented an array of 10 counting semaphores using sleep() and wakeup() with system calls sem_init, sem_wait, and sem_signal.
- Synchronization: Protecting the counter with semaphores fixes race conditions, giving an exact final count of 300,000.
- Producer-Consumer: Managed a 5-slot bounded buffer using mutex, empty, and full semaphores to prevent buffer overflow and underflow.

3. Q3 — Deadlock Detection & Avoidance

Design & Results

- Matrix Detection: detect_deadlock() uses resource allocation matrices to spot processes trapped in a circular wait.
- Banker's Algorithm: Checks if allocating resources keeps the system in a safe state before granting requests.
- Pipe Deadlock Simulation: Simulated an actual deadlock using two processes waiting on cross-dependent pipe descriptors, which detect_deadlock() successfully flagged.

4. Q4 — Priority Preemptive Scheduler with Aging

Design & Results

- Priority Scheduling: Replaced Round-Robin scheduling with a priority-based model ($1 = highest, 20 = lowest$, default $= 10$). Added set_priority and get_priority system calls.
- Aging Policy: If a process waits over 30 ticks without running, its priority number decreases by 1 to prevent starvation.

- Result: Higher priority tasks finish first (~100–110 ticks vs. ~300 ticks), while low-priority tasks still complete thanks to aging.

5. Key Problems Faced & Solutions

1. Thread Silent Failure: File descriptors were not copied initially; added filedup() in clone() so printf works in threads.
2. Lost Signals: Wrapped semaphore decrement checks inside while (value <= 0) loops to handle broadcast wakeups safely.
3. Pipe Deadlock Cleanup: Issued kill() to deadlocked pipe processes so readers catch the signal and exit cleanly.
4. Starvation: Implemented the 30-tick aging rule to ensure lowest-priority processes eventually run.

6. Conclusion

Successfully implemented kernel threads (clone/join), counting semaphores, deadlock detection/avoidance, and a preemptive priority scheduler with aging in xv6.